Question	Answer	Marks
5(a)(i)	potential difference per unit current	**B1**
5(a)(ii)	resistor connected to cell in a closed loop **and** correct circuit symbols used for all components	**B1**
	ammeter connected in series with resistor **and** voltmeter connected in parallel with resistor	**B1**
5(b)(i)	(resistance) = 1.38 / 0.276 = 5.00 (Ω)	**A1**
5(b)(ii)	$\rho = RA / L$	**C1**
	$\rho = 5.00 \times \pi \times (0.496 \times 10^{-3})^2 / (4 \times 0.864)$	**C1**
	$= 1.12 \times 10^{-6}\ \Omega\ \text{m}$	**A1**
5(b)(iii)	calculation of a fractional or percentage uncertainty in one quantity 0.001 / 0.864 **or** 0.002 / 0.496 **or** 0.02 / 1.38 **or** 0.001 / 0.276	**C1**
	percentage uncertainty = $[(0.001 / 0.864) + (2 \times 0.002 / 0.496) + (0.02 / 1.38) + (0.001 / 0.276)] \times 100$ = 2.7%	**A1**
5(b)(iv)	$\Delta\rho = 0.03 \times 1.12 \times 10^{-6}$ $= 3 \times 10^{-8}\ \Omega\ \text{m}$	**A1**

Question	Answer	Marks